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Bihar Engineering University, Patna

End Semester Examination - 2023 Semester-IV

Time: 03 Hours

Code: 104401 ***Subject: ECE Analog Circuit*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Write the answer of the following (Any seven question only): **[14]**

(a) Which of these are incorrect about Darlington amplifier?
- (i) It has a high input resistance
- (ii) The output resistance is low
- (iii) It has a unity voltage gain
- (iv) It is a current buffer

(b) What is a cascade amplifier?
- (i) A cascade of two CE amplifiers
- (ii) A cascade of two CB amplifiers
- (iii) A cascade of CE and CB amplifiers
- (iv) A cascade of CB and CC amplifiers

(c) What is the output of a class B amplifier for sinusoidal input?
- (i) Sinusoidal amplifier
- (ii) Half-sinusoidal
- (iii) Sinusoidal with higher frequency
- (iv) Square wave

(d) In a class B amplifier, it is found that DC power is 25W, find the ac power.
- (i) 10 W
- (ii) 62.5 W
- (iii) 25 W
- (iv) 50 W

(e) In an oscillator if phase of feedback is same as that of oscillation waveform then feedback is called
- (i) Positive feedback
- (ii) Negative feedback
- (iii) Cannot be predicted
- (iv) Either positive or negative depending upon frequency

(f) For a MOSFET, the pinch-off voltage is -3V. Gate to source voltage is 20V. W/L ratio is 5. Process transconductance parameter is $40\mu A/V^2$. Find drain to source current in saturation.
- (i) 0.10 mA
- (ii) 0.05mA
- (iii) - 0.05mA
- (iv) -50A

(g) For a MOSFET, the pinch-off voltage is -3V. Gate to source voltage is 20V. W/L ratio is 5. Process transconductance parameter is $40\mu A/V^2$. Find drain to source current in saturation.
- (i) 1100 Ω
- (ii) 11000 Ω
- (iii) 900 Ω
- (iv) 110 Ω

(h) The hybrid π model is used for
- (i) Analysis of BJT for low frequency
- (ii) Analysis of BJT at high frequency
- (iii) Analysis of BJT for feedback
- (iv) Analysis of BJT for large signal

(i) Which oscillator is characterized by a split capacitor in its tank circuit?
- (i) RC phase shift oscillator
- (ii) Colpitts oscillator
- (iii) Wien bridge oscillator
- (iv) none

(j) The gate voltage in a JFET at which drain current becomes zero is called
- (i) pinch-off voltage.
- (ii) saturation voltage
- (iii) active voltage.
- (iv) cut-off voltage.

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Q.2 Solve this question : **[14]**

Explain construction and working of Enhancement and Depletion type MOSFET.

Q.3 Solve this question : **[14]**

What is power amplifier. Explain Transformer coupled Class-A amplifier.

Q.4 Solve this question : **[14]**

Barkhausen criteria play an important role while designing Oscillator justify. Explain Hartley and RC phase shift Oscillator.

Q.5 Solve this question : **[14]**

Explain the effect of voltage series and voltage shunt feedback on gain & impedance of amplifier.

Q.6 Solve this question : **[14]**

The BJT is connected as a CE amplifier. Determine Z_i, Z_0, A_v, A_i hybrid model.

Q.7 Solve this question : **[14]**

Describe current mirror circuit in BJT. What are the condition to design current mirror circuit. Obtain the expression for current transfer ratio.

Q.8 Solve this question : **[14]**

Derive the expression for voltage gain and current gain and B.W of multistage amplifier.

Q.9 Solve this question : **[14]**

In brief explain the working of voltage series and current series circuit. Explain the same in case of BJT.

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